Design of traffic signal method

(4) Is code method

- based on?

- Red time = ?+?

- Amber time ko green time ke bad kyo dete

- stop to go
- stop

(4) Is code mthd

It is based on combination of approximate methods & webster's method

Red time : green time + amber time

Amber time is provided after green time because-

(1) to allow the traffic ~~is~~ change from stop to go.

(2) to allow sufficient time to stop vehicle approaching the intersection beyond SSD.

Traffic signs.

- Types

Traffic signs

- Regulatory or mandatory sign
- informatory sign
- cautionary or warning sign.

12 Traffic signs

- stop sign
  - type
  - shape, colour

Stop sign is a regulatory sign.
with octahedral shape
Red background white text

- Giveway
  - shape
  - type

Giveway △ular sign with apex/x down

it is also a Regulatory sign.

(2) - informatory signs are provided for

(2) Informatory signs are provided to give information only

- marking
- jn approch

- route marking
- road junction approach.

(3) - Warning or cautionary sign
  - if not followed then causes

(3) Warning or cautionary sign
• if not followed then causes fatal accident

Class 10 14 Aug 2024

Pavements
- Types
  - flex,
  - rigid

flexible pvmt
- it consits of what at the surface
- it is based on (method of action)
- flexural str of this pvmt is..
- layers
  - s
  - b
  - sg
  - st

Highway pavement design

Types of pavement

1) flexible
2) rigid

flexible pavement
- it consicts of a series of layers with highest quality material @/or near the surface.
- it is based on function of load distribution through grain to grain action
- flexural strength or bending strength of this pavement is zero or negligible
- it consits of 4 layers

surface or wearing course
base course
subbase
subgrade

S10-2

Types of roads
hai flexible pvmt
- approx age
- IRC code

Rigid pavement
- Top layer kis kis h
- kasi load transfer kota h?
- kya major factor for design
- kaise failure kojeet leta h?
- kitne layers aur kya kya
- IRC
- approx life?

- approx age 15 yrs (Range 15)
- IRC 37-2012

Rigid pavement

In rigid pavement top layer is cement/cr cement concrete slab
- it can transfer load by slab action.
- its flexural/bending strength is major factor for design.
- it is able to bridge over localised failures.
- It is 3 layered system.

Surface/wearing Course
base
subgrade

- IRC 58:2002
- approx life = 30 yrs & above

flexural str
- ength
  - Definition
  - fcr formula

- flexural ka matlab hota h bending strength
- $f_{cr} = 0.7\sqrt{f_{ck}}$

10.3

Semirigid or
- Semiflexible
  - Design factors
    - Design life
    - Defnn
    - Anticipated traffic
      - formula
        - A formula
        - P
        - r
        - n

Semirigid / Semiflexible pavement

Design factors

1) Design life

The number of standard axels that can be carried before strengthening of the pavement is called design life.

2) Anticipated traffic.

$$A = P[1 + r]^n$$

A is Design traffic intensity (Volume) = no of commercial vehicle / Day

P is number of commercial vehicles per day counted on last count

r is growth rate of traffic

n is number of yrs between last count & till the end of life of pavement

162

10.1 Designations of Semi flexible pavement

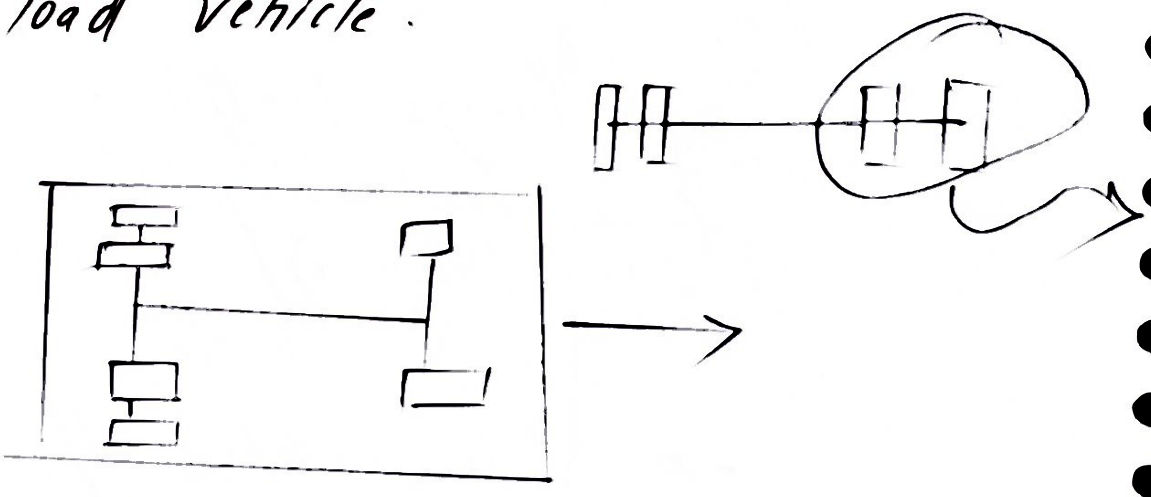
- (3) Design wheel load
  - Defn of max load (what is max wl load)
- (4) Equivalent Single wheel load
  - loaded vehicle ko maximum wheel load limit ke andr rkhne ke liye kya karna padega
  - diagram
    - distance
    - load ref

3) Design wheel load (maximum wheel load)

Total load that influences the thickness requirement of pavement is called maximum load

4) Equivalent single wheel load

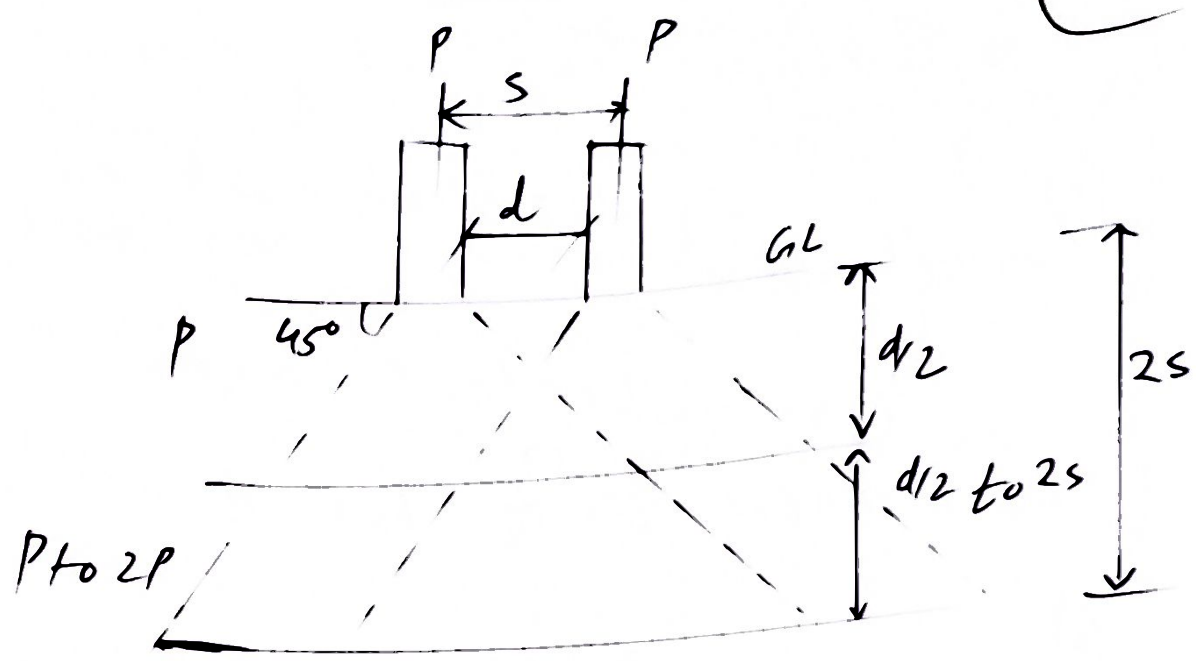
. To maintain maximum wheel load within the specified limit and to carry a greater load it is necessary to provide dual wheel assembly to the rear axel of load vehicle.



dual wheel rear assembly

ESWL

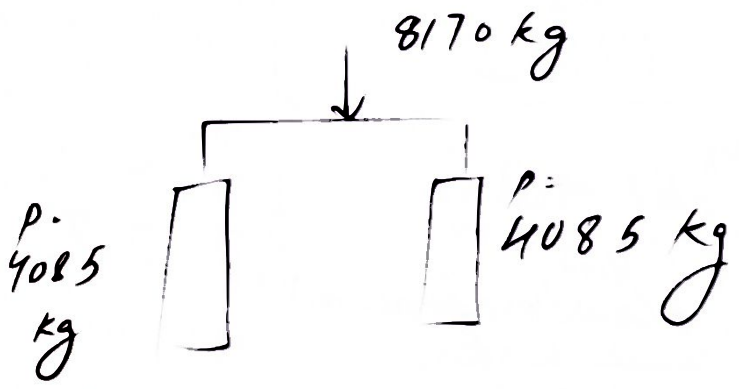
ESWL diagram



- load distance table

distance	load (ESWL)
d/2	P
d/2 < d < 2S	P to 2P
d ≥ 2S	2P

- P values



- d kya h?

d: clear distance or inner distance between wheels.

- S kya h?

S: centre to centre spacing between wheels

Design factors to be considered

5) Contact pressure
   - = ?
   - contact area shape?

6) Tyre pressure
   - alternate name

7) Rigidity factor
   - formula
   - Taor C kivalye kya hogi when
     - RF = 1
     - RF ≠ 1
       - name
     - RF > 1
       - name

- 7 Min
  - Mpa
  - N/mm²

5) Contact pressure
= load on wheel

Contact area is assumed as circle but it is mostly ellipse

6) Tyre pressure = inflation pressure

7) Rigidity factor = $\frac{\text{Contact pressure}}{\text{Tyre pressure}}$

It is the ratio of contact pressure to tyre pressure (RCT)

RF = 1	C.T = 7 kg/m²
RF < 1 (:: High tyre pressure)	C < 7 kg/cm², T > 7 kg/cm²
RF > 1	C > 7 kg/cm², T < 7 kg/cm²

(:: low tyre pressure)

0.7 Mpa

0.7 N/mm²

7 kg/cm²

flexible pavement

Empirical mthd.
- Based on
- Methods(4)
  - G
  - C
  - M
  - S

① Empirical methods.

• Based on physical properties or strength of soil subgrade.

• Methods

└ GI, CBR, Mcleod, Stabilometer

② Semiempirical
- Method

② Semiempirical/semitheoretical methods.

• Triaxial method

③ Theoretical
- method

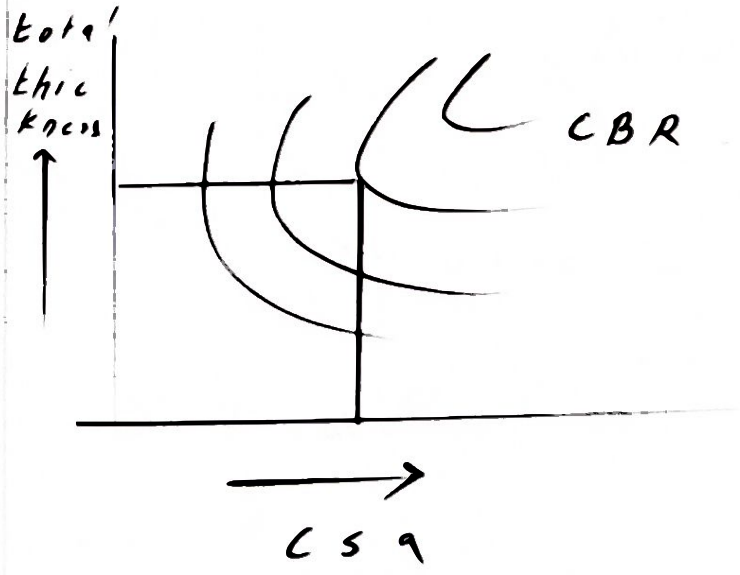
③ Theoretical method

• Burmisters method.

CBR method
- based on?
- graph

CBR Method

IRC/ recommends method based on CBR & cumulative standard axle load



10.9

cum standard axle formula?

Cumulative Standard Axle : $\frac{365 \times [(1+rate)^{life} - 1]}{rate}$ x at present no of commercial vehicles in nsa x lane distrib^n factor x vehicle damage factor

CBR design worksheet me diyah

worksheet 2 - 14 Aug 2024.

minimum bample size for axle load survey table

* impsecn

minimum sample size for axle load survey	Total number of commercial vehicles per day	% min of commercial vehicles surveyed / sample
- < 3000	< 3000	20%
- 3000 - 6000	3000 - 6000	15%
- > 6000	> 6000	10%

Equivalency factor
- kya krta hai
- formula

Equivalency factor

it is a factor that converts actual axle load into standard axle load

equivalency factor EF $= \left[\frac{\text{actual axle load}}{\text{standard axle load}}\right]^4$

Triaxial method
- thickness of pavement is depen dent h/our formula.
  - E_1, E_2
  - T_1, T_2

Triaxial method (semiempirical)

Thickness of pavement required is based on 2 layer systems.

$$\frac{T_1}{T_2} = \left(\frac{Ep_1}{Ep_2}\right)^{1/3}$$

Ep_1, Ep_2 are modulus of elasticity of pavement material for layer 1 and layer 2.

T_1 & T_2 are thickness of layer 1 & layer 2

Burmisters mtd
- alternate name

Burmisters method [theoretical]

- layered system

C. Res val method
- based on which two constants
- type of mthd
- formula

California resistance value mthd

- This method is based on two constants:-
  1) Stabilitometer value R
  2) cohesiometer value C
- empirical method
- formula $\left(\frac{T_1}{T_2}\right) = \left(\frac{C_2}{C_1}\right)^{1/5}$

Conclusions
- CBR kya hai ~~imp~~
- Max CBR
- GI rank/range
  - Good rank
  - poor rank

Conclusions of flexible pavement design

- CBR is a - measure to indicate relative strength of paving material
- Maximum value of CBR of any material - 100
- Group index range 0 to 20
  (0 → Good, 20 → poor)

Conclusions

- Can GI be -ve

- Group Index cannot be negative if comes out to be -ve, report it 0.

#Vimp

Q Which of the following property is true or correct for soil stabilisation.

- kab subgrade treat kreage
- base course I_p after treatment
- Subbase I_p after treatment 2001, 2008

2017 impqn.

Ans ⓐ subgrade should be treated if it has a soaked CBR 5 or less

ⓑ base course after treatement Should have plasticity inder $I_p \leq 2$.

& CBR in excess of 100.

ⓒ Sub base after treatement should hav CBR $I_p \leq 5$

QIRC 37 ——— recommendation for CBR method.

ESE 2022

- CBR kis soil pe kratehi
- Soil ko kahan tak compact kratch
- CBR test sample kitne din soak kratch

[Flexible pavemen design]

- CBR test should be performed on Remoulded soil on the field
- for the design of new roads the subgr ade soil sample should be compacted @ OMC to prot tor density
- for new construction the CBR test sample may be soaked in water for 4 day period before testing

Bitumen tests
- flash & fire pt test
  - kya test h ye
  - Apparatus
  - flash pt
    - Defn
    - temp
  - fire pt
    - Defn
- Solubility test
- Spot test
  - kiske liye h

flash & fire pt test

(13.6)

- These are primarily safety tests.
- Pensky martin closed cup & open cup apparatus
- flash point:
The lowest temperature @ which the vapour of a substance momentarily takes fire in form of flash
- temp 175°C
- fire pt: the lowest temp at which material gets ignited & burns under specified condition of test.

Solubility test

Spot test

- Test for detection of overheated or cracked bitumen

Bitumentests

- flash & firept test
  - kya testh ye
  - Apparatus
  - flashpt
    - Defn
    - temp
  - firept
    - Defn
- Solubilitytest
- Spot test
  - kiske liye hai

flash & firept test

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Solubility test

Spot test

- Test for detection of overheated or cracked bitumen

13.2

Bitumen tests
- Spot test
  - kinetech
- water content
  - Max wtr content in bitumen
- float test
  - to use
  - to test
  - to indicate
  - value
  - unit

naptha is used

Water content test

- Maximum water content in bitumen should not be less than 6.2% by weight

float test

- it is also a measure of consistency and viscocity.
- Higher the test value, stiffer the material
- measured in time taken in seconds

Grading of bitumen

Grading of bitumen

- In india bitumen is represented by penetration value, penetration grade. Eg 80/100, 60/70, 30/40.
- to represent

Grading of bitumen

- New method

Now a more rational method of grading is used by BIS — in terms of viscosity grading. (VG)

	Absolute Viscosity	kinematic viscosity	Penetration val
VG10 - kiemat viscosity / Penteration val	800 poise	250 centistoke	80-100
VG20	1600	300	60-80

- General application

General application

- VG 10 (80-100): spraying, paving applications in cold region
- VG 20 (60-80): paving application in cold region @ high altitude & northern part of india
- VG 30 (50-70): paving application most part of India
- VG-40 (40-60): used in high stressed area such as intersections toll plaza truck layby

13-9) grading of bit
- test temp
  - abs visc
  - kinemat visc
- TFOT
  - kya findkshi ye test?

absolute viscoit test temperature 60°C
kinematic viscocity test temper 135°C
penetrn test 25°C

Thin film oven test

• The different viscocity values are observed in two simmilar viscocity grades of bitumen which is foond by this test.

PMB --- full form ---- PMB polymer modified bitumen
CRMB ---- CRMB (rumbrubber modified bitumen

• khan use krenge PMB aur CRMB ko

These are used only in wearing coorse depends on requirement

Class 14 — 1 Aug 2024 (14.1)

Rigid pavement

- kya kya cheez hoti h is me?

Rigid pavement

- Consists of - a. cement concrete Wearing course & it is of 3 layer system.

- kitni layers h?

Subgrade
base
Surface course

Detailing worksheet me milega)

- Pavement failure

- kya kya failures h?

Pavement failure.

- Rutting
- Revelling
- Pot holes

} CC + damar road

- Craziness

- Aligator crack kisme dikhta h?

→ Aligator crack or map cracking.

Seen in CC pavement

14.2 Rig pvmt Design steps

- load trfer
  - its check
  - it keeps load carrying capacity
  - load carry stabilized
- formula
  - P kg-lb
  - S kg-lb
  - k kg/cm²
- Radius of rel stiffness
  - Measure of rel stiffness

Rigid pavement design steps

1. load transfer

load carrying capacity is mainly due to rigidity & high modulus of elasticity of the slab itself

$E_c = 5000\sqrt{f_{ck}}$

$k = \frac{P}{S}$ $\quad \boxed{P = kS}$

Pressure = Modulus of reaction × deflection

P is pressure below plate

S is deflection at the center of plate

k is modulus of subgrade reaction.

2 Radius of relative stiffness

A certain degree of resistance to the deflection is offered by the subgrade hence relative stiffness 'l' is defined by Westergaard for detailing deformation characteristics

Design steps Rigid pvmt

2. Radius of relative stiffness (otd.

- formula.

$$\ell = \left[\frac{E_c h^3}{12k(1-\mu^2)}\right]^{1/4}$$

$E_c = 5000\sqrt{f_{ck}}$

$$\text{Radius of Relative stiffness} = \left[\frac{\text{Modulus of elasticity of concrete} \times (\text{slab thickness})^3}{12 \times \text{Modulus of sub grade reaction} \times (1 - \text{poissons ratio of conc } (0.15))^2}\right]^{1/4}$$

- units kg/cm²

- E
- h
- k
- ℓ

E_c is in kg/cm²
h is in cm
k is in kg/cm³

} ℓ is obtained in cm.

3. Equivalent load of resisting secn

- formula when $a < 1.724h$

3. Equivalent radius of resisting section (b)

(i) $b = \left[\sqrt{1.6a^2 + h^2} - 0.675h\right]$ when $a < 1.724h$

- formula when a is different.

(ii) $b = a$ when (i) is not satisfied

(Generally yahi use hota h)

- a kya h?

a is radius of wheel load distrib?

- formula

$$a_{cm} = \sqrt{\frac{P \text{ kg/cm}^2}{q_{cm}\pi}}$$

- p kya h

- q kya h?

q is contact pressure
p is pressure below the plate.

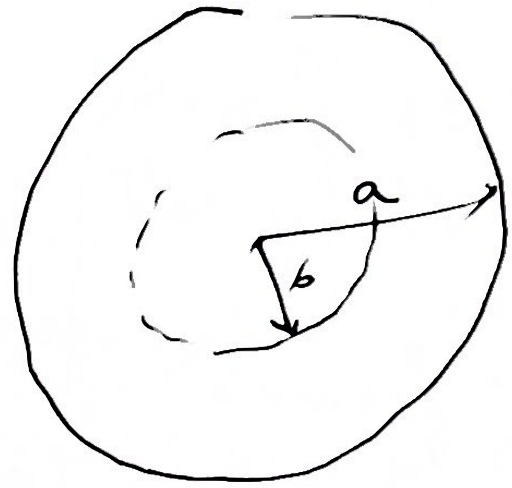
14.1) Compute the equivalent radius of resisting section of 15 cm slab if the ratio of radius of wheel distribution to thickness of slab is 0.5.

Ans:

b: ?

a/h = 0.5

h = 15 cm



a = 0.5 × 15 = 7.5 cm

1.724h = 1.724 × 15 = 25.86 cm.

condition check

a < 1.724 h

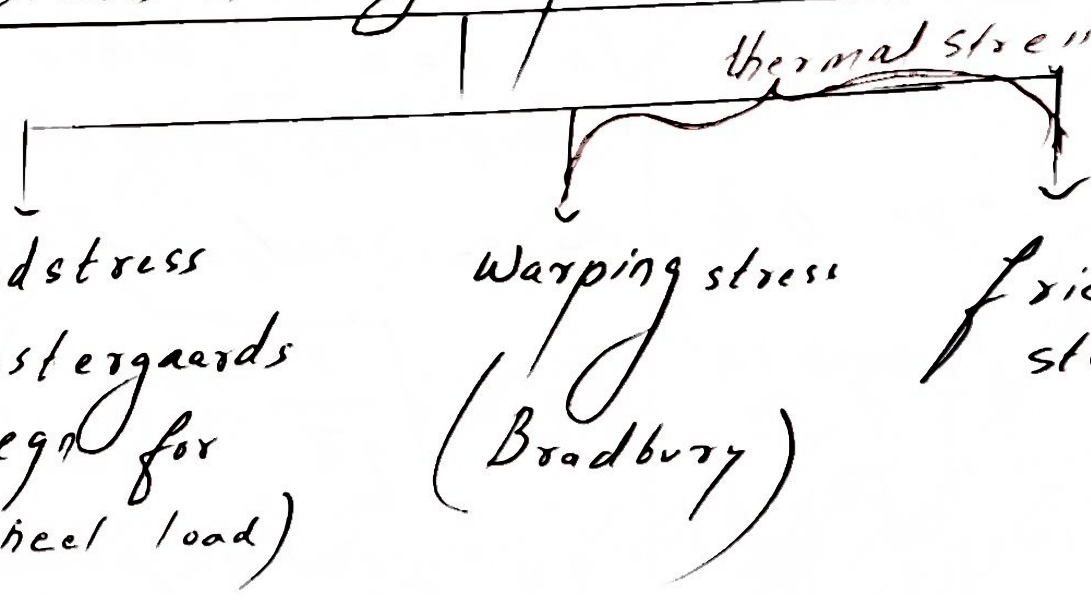
7.5 cm < 25.86 cm

$b = \sqrt{1.6a^2 + h^2} - 0.675h$

$= \sqrt{1.6 \times 7.5^2 + 15^2} - (0.675 \times 15)$

= 7.62 cm

Stresses in rigid pavement



Design of rig pvmt
Stresses in pvmt

kya kys stresses h usm avg krungi thermal h.

Design of rigpvmt critical regions for rigid pavement are

for rigid pavement critical regions are -

1. corner
2. edge
3. interior

- Westergaard modified westergaard smethy. farkhaz

In westergaards theory load stresses are given for interior edge, corner region.

In modified westergaards the load stresses are given for edge and corner region only.

- formula for corner loading

Corner loading stress $\sigma_c = \frac{3P}{kg/cm^2\ h^2}\left[1-\left(\frac{h\sqrt{2}}{l}\right)^{3.2}\right]$

14.6

Stresses in rig pvmt

- thermal stress
  - Warping stress formula @
    - (i) interior
      - α
      - C_x, C_y
      - x
      - t
    - (ii) edge
      - formula

Thermal stress

Warping stress (bradbury)

(i) interior

$$\sigma_w = \frac{E \alpha t}{2}\left[\frac{C_x + \mu C_y}{1-\mu^2}\right]$$

α is coefficient of thermal expansion per degree centigrade. $1^{\circ}C$

C_x & C_y are coefficients in x & y direction.

x is shorter dimension y longer dim

t is temp diff during day 6th top & bottom layer of slab. $^{\circ}C$.

(ii) $\sigma_{we} = \frac{C_x E \alpha t}{2}$

$\frac{C_y E \alpha t}{2}$ } more

thermal stress important
Warping stress
@ (iii) corner
formula

$$\sigma_{wc} = \frac{E\alpha t}{3(1-\mu)}\sqrt{\left(\frac{a}{l}\right)}$$

frictional stress
frictional stress is always zero @ corner kyonki corner pe kaaran slab ka soil se contact khatam ho jata h/ aur corner lift ho jata h jisse corner pe frictional stress dere top nahi hota.

frictional stress induced in bottom fiber due to contact with soil/subgrade should be equal to force in cc pavement due to movement.

(for $\rightleftharpoons$)

Force in cc pavement = frictional resistance due to subgrade

- frictional stress
- corner pe kitna h?
- kyon corner pe wob hota h jo hota h?
- $\rightleftharpoons$ ke liye kya = hone chahiye
- rig pvmt me friction kerespectme.

- 14.8 Thermal stress in rigpvmt
  - frictstress
    - formula for =
      - 2400 kg / m^3
      - kya hai

$$\sigma_f(B*h) = [B \times h \times \frac{L}{2}]\gamma_c = f$$

$$\sigma_f = \frac{S_c f L}{2}$$

→ 2400 kg/m^3

σ_f kg/m^2 unit stress or frictional stress developed in cement concrete

$$= \frac{\text{mass density} \times \text{friction coeff (1.5)} \times \text{Slab length cm}}{2}$$

B is slab width cm

h is slab thickness cm

Critical combinn of stresses.
- kya kya combos hai
  - 5 nd ay
    - Critical stress kahan?

Critical combination of stresses

1. Summer midday
2. Winter midday
3. Winter midnight

① Summer midday

critical stress is defined for edge region

Critical combinn of stresses:

- ① Summer midday
  - formula
- ② Winter midday
  - Crit region kabhi
  - formula
- ③ Winter midnight
  - Crit region kabhi
  - formula

$\sigma_{critical} = \sigma_{load} + \sigma_{warping} - \sigma_{friction}$

② Winter midday

Crit region edge region.

$\sigma_{critical} = \sigma_{load} + \sigma_{warping} + \sigma_{friction}$

③ Winter midnight.

for winter midnight critical combination of stress obtained

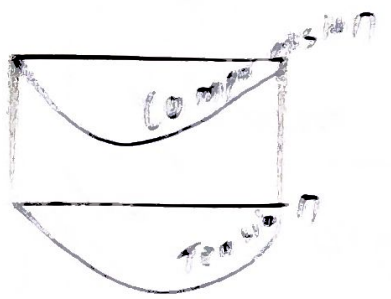
@ corner region

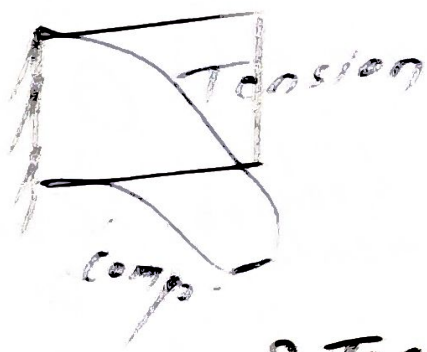
$\sigma_{crit} = [\sigma_{load\ corner} + \sigma_{Warping\ corner}]$

- Stress pattern in load
  - Stress
    - 1.1 distribution
      - edge & interior
    - 1.2 distribution
      - corner
- Warping stress.
  - 2.1 Daytime me slab kaisa dikhta h? Top pe ky ata h?
  - 2.2 Nighttime kaisa dikhta h? kya stress atah?

Stress pattern compression or tension.

load stress



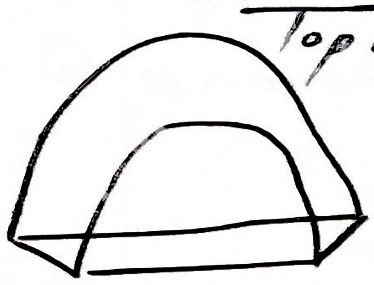


edge → Interior → Tension (bottom)

corner → Tension (top), Compression (bottom)

Warping stress (for same edge corner & interior)

Due to warping stress @ daytime slab looks like →



@ nighttime →

